

Many-Source Parity Networks

Registered Ablation Signatures in Fused Gravitational Scenes

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Abstract

The gravitational coherence surface (GCS) formalism identifies contribution–context parity in a declared gravitational scene. This paper develops the corresponding many-source construction after replacing assumed source-owned scalar fields with registered tensor ablation. A full fused scene is solved; each declared source is ablated in turn; every comparison scene is re-solved and registered to a common support; and the resulting source-conditioned change in the electric part of the Weyl tensor is retained as the primary evidence object. For source i ,

$$\Delta_i E_{ab} = E_{ab}^{(\text{full})} - \tilde{E}_{ab}^{(-i)}.$$

The Frobenius amplitude $S_i = \|\Delta_i E\|_{F,\text{ref}}$ is adopted as the default direction-agnostic scalarization. Pairwise equality $S_i = S_j$ defines a candidate signature-parity set. The retained many-source network consists of those portions on which the tied signatures are also locally dominant over the remaining source-conditioned amplitudes. This yields source-conditioned signature cells, pairwise faces, triple seams, and higher-order nodes without pretending that the fused geometry decomposes uniquely into private source fields. The phrase “source-readable” is retained only as interpretive shorthand.

The legacy weak-field proxy $Q_i \propto M_i/r_i^3$ is retained as an analytic benchmark lane. It recovers midpoint planes for equal masses and the deformation law $r_i/r_j = (M_i/M_j)^{1/3}$ for unequal masses, but it no longer supplies the primary ontology. A second network object records how a fixed registered relational complex changes under ablation, producing removed, born, and relabeled strata as a source-signature fingerprint. Neither network is a force boundary, material partition, causal horizon, orbital-stability limit, or modification of general relativity. They are declared diagnostic geometries of source-conditioned readability in one shared field.

Keywords: general relativity; electric Weyl tensor; trace-free tidal tensor; tensor ablation; Frobenius norm; many-source parity network; relational signature; dominance complex; gravitational coherence surface.

1. Introduction: from pairwise proxies to fused-scene signatures

The founding many-source construction asked a simple question: where does one source-readable gravitational contribution become equal to another? In a weak-field point-mass lane, assigning

$$Q_i(x) \propto \frac{M_i}{r_i(x)^3} \tag{1}$$

made that question analytically transparent. Pairwise equality produced midpoint planes for equal masses, mass-weighted deformations for unequal masses, and seams where three or more equality surfaces met.

That scaffold remains useful, but its interpretation must mature with the GCS program. In a fused nonlinear geometry, general relativity does not ordinarily provide a unique decomposition into independently owned source fields. Even in exact time-symmetric conformal initial data, the metric and curvature depend nonlinearly on the combined conformal factor. A source-source network should therefore not begin by assuming that each source already possesses a separable scalar field inside the full scene.

The revised construction begins with intervention rather than ownership. Each source is removed from the roster in turn, the scene is re-solved, and the resulting geometry is registered against the same full-scene support. The source-conditioned response is tensorial. Only after that tensor bundle is built is a scalar amplitude used to extract one network.

The question becomes:

Where do two registered source-conditioned responses reach equal strength, and which equality strata actually organize the local many-source signature structure of the fused scene?

The answer is a registered ablation-signature network. It preserves the original paper’s faces, seams, and junctions while replacing its scalar-first ontology with an operational GR-native method.

2. Scope and hierarchy

This paper defines two related but distinct many-source objects.

1. The *signature-dominance network* compares the default Frobenius amplitudes of registered source-ablation tensors. It partitions the declared active support into source-conditioned signature cells and their shared parity strata.
2. The *ablation-complex difference* records how a fixed registered relational skeleton changes when one source is removed. It retains removed, born, and relabeled cells, faces, seams, and nodes as an ordered source-signature fingerprint.

The tensor field is primary in both constructions. The Frobenius amplitude is the default scalar read, not the whole evidence object. Spectral norms, eigenvalues, eigenframes, invariant polynomials, degeneracy proximity, and trajectory responses remain secondary channels.

The paper does not claim that these networks are material territories, unique invariants, or automatic observables. It does not replace the metric, curvature tensors, geodesic deviation, or numerical relativity. It supplies a disciplined diagnostic layer over solved scenes.

3. Registered many-source ablation bundle

3.1 Declared scene and intervention family

Let $(\Sigma, \gamma_{ab}, K_{ab})$ denote the full declared spatial scene containing sources $\{S_i\}_{i=1}^N$. The declaration includes the physical regime, source roster, construction or solver, units, coordinates, domain, masks, datums, and comparison conventions.

For each source i , let $(\Sigma_{-i}, \gamma_{ab}^{(-i)}, K_{ab}^{(-i)})$ denote the scene obtained by removing that source from the roster and re-solving under the same declared method. Ablation is counterfactual. It is not physical destruction, radiation, or time evolution.

Table 1: Hierarchy of the revised many-source construction.

Object	Role
Full solved scene	One fused geometry containing all declared sources.
Registered ablation bundle	One tensor response $\Delta_i E_{ab}$ for each declared source intervention.
Default signature amplitude	$S_i = \ \Delta_i E\ _{F, \text{ref}}$, evaluated on one common registered support.
Source-readable cell	Region in which one signature amplitude is maximal under the declared comparison.
Parity face	Dominant tie set between two source-conditioned amplitudes.
Seam or node	Dominant tie among three or more amplitudes, with dimension set by transversality.
Complex difference	Removed, born, and relabeled relational strata under a fixed-skeleton ablation comparison.

3.2 Common registration

Tensor subtraction across different solved geometries is not self-defining. For each intervention, declare a point identification

$$I_i : U \subset \Sigma \longrightarrow U_{-i} \subset \Sigma_{-i} \quad (2)$$

and a frame map

$$P_i(x) : T_x \Sigma \longrightarrow T_{I_i(x)} \Sigma_{-i}. \quad (3)$$

The registered ablated electric-Weyl tensor on the full-scene support is

$$\tilde{E}_{ab}^{(-i)}(x) = P_a^c(x) P_b^d(x) E_{cd}^{(-i)}(I_i(x)). \quad (4)$$

All source interventions must be compared on the same declared support and under compatible frame and reference-metric conventions. Otherwise pairwise amplitude equality is not meaningful.

3.3 Electric Weyl witness and source-conditioned response

Let n^μ be the unit normal to the slice and $C_{\mu\alpha\nu\beta}$ the Weyl tensor. The electric part

$$E_{\mu\nu} = C_{\mu\alpha\nu\beta} n^\alpha n^\beta \quad (5)$$

records the trace-free free-gravitational tidal structure relative to that normal [6, 9]. In vacuum, where Riemann equals Weyl, its spatial part is the complete tidal operator governing geodesic deviation relative to the chosen normal. Inside matter, Ricci curvature also contributes to the full geodesic-deviation operator, so the witness and matter assumptions must be declared.

For each source i , define the registered relational signature tensor

$$\Delta_i E_{ab}(x) = E_{ab}^{(\text{full})}(x) - \tilde{E}_{ab}^{(-i)}(x). \quad (6)$$

This is the response of the whole solved scene to the presence of i , including altered nonlinear cross-terms. It is not asserted to be a privately owned field of that source.

4. Default Frobenius signature network

4.1 Signature amplitudes

Let γ_{ref} be one declared positive-definite reference metric on the registered support. For a covariant tensor A_{ab} define

$$\|A\|_{F,\text{ref}} = \left(\gamma_{\text{ref}}^{ac} \gamma_{\text{ref}}^{bd} A_{ab} A_{cd} \right)^{1/2}. \quad (7)$$

The default source-conditioned signature amplitude is

$$S_i(x) = \|\Delta_i E(x)\|_{F,\text{ref}}. \quad (8)$$

The Frobenius norm is rotation-invariant within the fixed physical inner product and reads the full tensor amplitude without selecting a preferred eigenvector. It is a default scalarization, not an assertion that direction and spectral shape are unimportant.

4.2 Candidate pairwise parity sets

For every pair (i, j) define the residual

$$\Psi_{ij}(x) = S_i(x) - S_j(x) \quad (9)$$

and the raw signature-parity set

$$\Sigma_{ij} = \{x \in U : \Psi_{ij}(x) = 0\}. \quad (10)$$

The sign records which registered source-conditioned response has the larger default amplitude:

$$\Psi_{ij} > 0 \Rightarrow S_i > S_j, \quad \Psi_{ij} < 0 \Rightarrow S_j > S_i. \quad (11)$$

This is not a statement about force direction, acceleration, orbital capture, or ownership of space.

4.3 Dominance cells and retained faces

Raw equality sets proliferate as $N(N-1)/2$. Most do not organize the local source-conditioned structure because a third signature may dominate both tied amplitudes. The retention rule should therefore be part of the mathematics rather than an informal visual judgment. Let $U_{\text{act}} \subseteq U$ denote the declared active support after masks and any amplitude-confidence floor have been applied. Points outside U_{act} are unclassified rather than assigned by numerical noise.

Definition 1 (Signature-dominance cell). *The closed dominance cell of source i is*

$$\mathcal{C}_i = \{x \in U_{\text{act}} : S_i(x) \geq S_k(x) \text{ for all } k\}. \quad (12)$$

Its strict interior is

$$\mathcal{C}_i^\circ = \{x \in U_{\text{act}} : S_i(x) > S_k(x) \text{ for all } k \neq i\}. \quad (13)$$

The strict interiors are disjoint. Closed cells overlap exactly on dominant ties, so tie points are retained as boundary strata rather than assigned arbitrarily.

Definition 2 (Retained pairwise face). *The closed retained tie set between sources i and j is*

$$\overline{\mathcal{F}}_{ij} = \{x \in U_{\text{act}} : S_i(x) = S_j(x) \geq S_k(x) \text{ for all } k\}. \quad (14)$$

Equivalently, $\overline{\mathcal{F}}_{ij} = \Sigma_{ij} \cap \mathcal{C}_i \cap \mathcal{C}_j$. Its ordinary face stratum excludes higher-order ties:

$$\mathcal{F}_{ij}^{(2)} = \{x \in U_{\text{act}} : S_i(x) = S_j(x) > S_k(x) \text{ for all } k \notin \{i, j\}\}. \quad (15)$$

Thus triple and higher ties lie in the closure of several pairwise faces but are typed separately as seam or node strata.

The default many-source signature network is

$$\mathcal{N}_{\text{sig}} = \bigcup_{1 \leq i < j \leq N} \overline{\mathcal{F}}_{ij}. \quad (16)$$

This converts the original relevance-filtered union into a reproducible dominance complex. It resembles a weighted Voronoi construction in organization, but the weights are not prescribed distances or masses. They are scene-dependent amplitudes of registered whole-scene responses.

4.4 Seams and nodes

For a source index set $A = \{i_1, \dots, i_m\}$, define a dominant m -fold tie stratum by

$$\mathcal{J}_A^{(m)} = \{x \in U_{\text{act}} : S_{i_1} = \dots = S_{i_m} > S_k \text{ for all } k \notin A\}. \quad (17)$$

At a point of $\mathcal{J}_A^{(m)}$, choose $m - 1$ independent residuals, for example $S_{i_1} - S_{i_q}$ for $q = 2, \dots, m$. If their gradients are linearly independent, then in three dimensions the implicit function theorem gives a local stratum of dimension $3 - (m - 1)$. Thus generic pairwise ties are surfaces, triple ties are curves, and fourfold ties are isolated points. Degeneracy, symmetry, tangency, or loss of transversality can change this dimension and must be reported rather than visually regularized away.

5. Local existence, regularity, and branch discipline

Proposition 1 (Local pairwise parity surface). *Let S_i and S_j be continuously differentiable in an open region $U \subset \mathbb{R}^3$. If $x_* \in U$ satisfies*

$$S_i(x_*) = S_j(x_*) \quad (18)$$

and

$$\nabla(S_i - S_j)(x_*) \neq 0, \quad (19)$$

then the raw parity set Σ_{ij} is a smooth embedded surface near x_ . If additionally $S_i(x_*) = S_j(x_*) > S_k(x_*)$ for all $k \notin \{i, j\}$, that neighborhood belongs to the ordinary retained face stratum $\mathcal{F}_{ij}^{(2)}$.*

Proof. Apply the implicit function theorem to the scalar residual $\Psi_{ij} = S_i - S_j$. The nonzero gradient condition guarantees that at least one coordinate derivative is nonzero, allowing the zero set to be represented locally as the graph of a differentiable function. $\square$

Proposition 2 (Generic dominant seam). *Let three signature amplitudes S_i, S_j, S_k be continuously differentiable. At a point where*

$$S_i = S_j = S_k \geq S_\ell \quad \text{for all } \ell, \quad (20)$$

if the gradients of $S_i - S_j$ and $S_i - S_k$ are linearly independent, then the dominant triple-parity set is locally a one-dimensional embedded curve.

These are local statements. They do not guarantee global closure, compactness, unique connectivity, or persistence under changes of registration, resolution, masks, scalarization, or scene roster. A computed network may contain disconnected faces, bounded cells, unbounded cells, pinches, contacts, births, deaths, and relabeling transitions.

6. Relation to the single-source GCS

For each source i , the revised GCS compares its signature amplitude with the registered amplitude of the scene in which it has been ablated. Define

$$C_i(x) = \|\tilde{E}^{(-i)}(x)\|_{F,\text{ref}} \quad (21)$$

and

$$\Sigma_{\text{GCS}}^{(i)} = \{x : S_i(x) = C_i(x)\}. \quad (22)$$

The single-source GCS and the many-source signature network answer different questions:

Table 2: Two parity geometries in the registered ablation framework.

	Equality	Question
Source GCS	$S_i = C_i$	Where does the response to source i reach parity with the geometry of the re-solved scene without it?
Signature face	$S_i = S_j \geq S_k$	Where do two source-conditioned responses tie while jointly dominating the other source signatures?

The overlay of the family $\{\Sigma_{\text{GCS}}^{(i)}\}$ with $\mathcal{N}_{\text{sig}}$ provides a richer scene atlas. A dominance face can pass inside or outside a source’s GCS because source-source parity and source-context parity are distinct relations. Neither object is where a signature ends.

7. Legacy weak-field proxy benchmark

The original point-mass construction remains valuable as an analytic control. On a Euclidean weak-field background, declare

$$Q_i(x) = k \frac{M_i}{r_i(x)^3}, \quad r_i(x) = \|x - x_i\|. \quad (23)$$

Pairwise equality gives

$$\frac{M_i}{r_i^3} = \frac{M_j}{r_j^3}, \quad \frac{r_i}{r_j} = \left(\frac{M_i}{M_j}\right)^{1/3}. \quad (24)$$

Equal masses therefore produce the perpendicular bisector plane. For $M_i > M_j$, the equality surface lies closer to the lower-mass source. Three equal masses at the vertices of an equilateral triangle produce pairwise bisectors meeting on the symmetry axis perpendicular to the triangle plane.

These formulas test extraction geometry, symmetry handling, and mass-sensitive deformation. They do not establish that M_i/r_i^3 is the unique source-readable scalar in a fused scene. The revised interpretation is:

Table 3: Status of the original and revised network lanes.

Lane	Primary quantity	Status
Legacy proxy	$Q_i = kM_i/r_i^3$	Analytic weak-field benchmark with source attribution imposed by the model.
Tensor-ablation network	$S_i = \ \Delta_i E\ _{F,\text{ref}}$	Preferred registered diagnostic; includes the whole-scene response and altered non-linear couplings.
Secondary tensor lanes	eigenvalues, spectral norm, eigenframes, invariants	Sensitivity and morphology channels; not automatically equivalent to FN parity.

The proxy law should survive because it gives the instrument a clean unit test. It should not survive as the public foundation.

8. Fixed-skeleton ablation-complex difference

The dominance network compares the source-conditioned signatures present in one full roster. A second operation asks how the relational topology of the scene reorganizes when a source is removed.

Let $\mathcal{K}^{(+i)}$ denote a registered relational complex evaluated with source i present and $\mathcal{K}^{(-i)}$ the corresponding complex after ablation. The comparison is lawful only if both are evaluated on a fixed datum skeleton or connected by an explicit cell-matching map. On a fixed skeleton, define

$$\mathcal{R}_i^- = \mathcal{K}^{(+i)} \setminus \mathcal{K}^{(-i)}, \quad (25)$$

$$\mathcal{R}_i^+ = \mathcal{K}^{(-i)} \setminus \mathcal{K}^{(+i)}, \quad (26)$$

for removed and born strata, and retain a relabeling field L_i for cells whose membership or adjacency changes without disappearing.

The ordered object

$$\mathfrak{D}_i = (\mathcal{R}_i^-, \mathcal{R}_i^+, L_i) \quad (27)$$

is an ablation-complex fingerprint under the declared partition rule. It can expose a global reorganization even when the high-amplitude tensor footprint is localized. It is not the tensor signature itself. The tensor bundle witnesses geometric change; the complex supplies a datum skeleton on which topological reorganization can be read.

This distinction preserves three levels:

1. $\Delta_i E_{ab}$: global registered tensor response;
2. S_i : default direction-agnostic amplitude;
3. $\mathfrak{D}_i$: fixed-skeleton relational reorganization.

9. Static fused-scene implementation lane

The construction can be instantiated in time-symmetric conformally flat initial data. In Brill–Lindquist form,

$$\psi = 1 + \sum_{i=1}^N \frac{m_i}{2r_i}, \quad \gamma_{ab} = \psi^4 \delta_{ab}, \quad K_{ab} = 0, \quad (28)$$

and the vacuum Hamiltonian constraint is satisfied away from punctures [3]. On this slice, the electric Weyl tensor is obtained from the spatial Ricci tensor under the vacuum constraint and adopted sign convention. Because curvature is computed from the combined conformal factor, the scene is fused and cross-term aware; it is not a sum of independently solved source curvatures.

Under the declared one-at-a-time ablation protocol, and assuming each comparison scene can be generated by the same solver without auxiliary intervention solves, a complete N -source network uses $N + 1$ solved scenes: one full scene and one ablation solve per source. The nominal solve count is therefore linear in source count. This is a protocol-level count, not a universal complexity theorem: nonlinear matter reconstruction, constraint repair, gauge fixing, registration, storage, scalarization, and topology extraction may dominate the practical cost. The method avoids pairwise re-solving: pairwise faces are derived after the N source-conditioned bundles are placed on the common support.

A publication-grade run should report:

- source roster, masses, positions, construction, units, and masks;
- full and ablated solve conventions;
- point and frame registration for every intervention;
- common reference metric and scalarization;
- grid or mesh resolution and puncture treatment;
- face, seam, and node extraction tolerances;
- dominance ties, active-support thresholds, and near-tie confidence;
- FN-versus-spectral and registration sensitivity;
- fixed-skeleton rules for any complex-difference claim.

The exact static lane establishes a controlled laboratory for the network. It does not establish dynamical persistence, radiation content, observational detectability, or universality across initial-data families.

10. Tensor channels and degeneracy discipline

A Frobenius network compresses each tensor signature to one amplitude. Important structure can survive outside that scalar partition. The companion bundle should retain at least:

- ordered eigenvalues and spectral-shape ratios;
- principal-axis directions and eigenframe rotation;
- spectral norm and invariant-polynomial checks;
- eigenvalue-gap or degeneracy-proximity fields;
- angular coherence and radial root inventories;
- geodesic-deviation responses where physically appropriate.

Eigenframe rotation is undefined at repeated eigenvalues and unstable near degeneracy. Large rotation angles should never be interpreted without the accompanying spectral gap. A thin seam of near-90° rotation may indicate an eigenvalue-ordering transition rather than a discontinuity in physical geometry.

Norm agreement is evidence of scalarization robustness in a declared scene. It is not proof

that all scalar networks are physically equivalent. The FN network remains the default because it is direction-agnostic, rotation-invariant under the chosen inner product, and uses the full tensor amplitude.

11. Evidence register and claim discipline

Table 4: Current status of the many-source network program.

Result	Status	Scope or promotion requirement
Legacy M/r^3 parity law	Analytic bench- mark	Exact only in the declared weak-field proxy lane.
Registered source-ablation bundle	Defined and imple- mented	Exact registration and solve convention must be declared per source.
FN signature amplitude	Default scalariza- tion	Requires one common positive-definite reference inner product.
Dominance cells and re- tained faces	Defined	Must test ties, resolution, masks, and scalarization sensi- tivity.
Triple seams and higher nodes	Locally character- ized	Global topology and persistence remain scene dependent.
Fixed-skeleton complex dif- ference	Implemented con- ceptually	Requires a common datum skeleton or explicit matching map.
Cross-context source classi- fication	Seed	No invariant identity classifier established.
Dynamical network persis- tence	Open	Requires evolved spacetime data and time-aware regis- tration.

The network does not modify general relativity or introduce a new force. It is not a material tessellation, causal boundary, trapping horizon, orbital-stability map, or automatic observable. A signature-dominance cell is not sovereign territory. A face is not a force wall. A seam is not necessarily a trajectory separatrix. A static network is not a history. Coordinate-component registration is not convention-free. The Frobenius norm does not exhaust tensor structure. Source ablation compares solved counterfactual scenes; it does not model physical disappearance.

12. Conclusion

The original many-source parity paper supplied a clean geometric scaffold: declare a source-readable scalar, compare sources pairwise, and retain the equality strata that organize a scene. Its analytic benchmarks remain valuable. Equal weak-field point masses produce midpoint planes, unequal masses obey the one-third-power deformation law, and three-source symmetry produces triple-parity structure.

The revised formulation changes what earns the right to participate in that network. A source-readable contribution is no longer presumed to be a privately owned M_i/r_i^3 field inside a fused geometry. Each source is operationally identified through registered ablation. The primary object is the whole-scene tensor response

$$\Delta_i E_{ab} = E_{ab}^{(\text{full})} - \tilde{E}_{ab}^{(-i)}, \quad (29)$$

and its Frobenius amplitude

$$S_i = \|\Delta_i E\|_{F,\text{ref}} \quad (30)$$

is the default direction-agnostic read.

Pairwise equality $S_i = S_j$ supplies candidate faces. On the declared active support, strict inequalities define disjoint cell interiors, while dominant equalities define shared boundary strata. The condition $S_i = S_j > S_k$ for all $k \in \{i, j\}$ defines an ordinary pairwise face; triple and higher equalities are retained separately as seams and nodes. Together their closures form a reproducible many-source signature complex. A fixed-skeleton before/after comparison then adds a second fingerprint: the relational strata removed, born, or relabeled when a source is ablated.

The field remains one seamless geometry. The network does not divide it into owned territories. It records how finite source-conditioned responses become equal, dominant, and topologically entangled under a declared registration, active-support rule, and scalarization. The old paper’s geometry survives. Its physics has been upgraded from assigned proxy fields to registered relational signatures.

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